Project Abstract

# UniGPT

*An AI Academic Copilot for Smart University Management*

## Team Information


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## 1. Problem Statement

Modern universities still rely heavily on fragmented systems and manual workflows for managing academic activities. Students frequently struggle to access important course materials, notices, assignment guidelines, attendance records, and academic policies because these resources are scattered across multiple disconnected platforms such as ERP systems, WhatsApp or Messenger groups, Google Classroom, spreadsheets, and paper-based records.

Faculty members face similar challenges, spending significant time on repetitive tasks such as attendance tracking, assessment creation, grading, feedback generation, and academic communication. Administrative departments also lack real-time institutional visibility, making decision-making slow and inefficient.

## 2. Proposed Solution

UniGPT is an intelligent, centralized academic platform powered by Artificial Intelligence. Rather than forcing students, faculty, and administrators to navigate multiple disconnected tools, UniGPT unifies all major academic workflows into a single platform with role-based experiences. The system provides one integrated ecosystem for academic communication, course management, attendance, assessments, document sharing, analytics, and AI-assisted decision support.

The core innovation behind UniGPT is its Retrieval-Augmented Generation (RAG) architecture. Unlike generic AI chatbots that may produce inaccurate or hallucinated responses, UniGPT grounds every answer in institution-approved documents — syllabi, policies, handbooks, and lecture materials. Every response includes source citations and confidence scores, making the system trustworthy and auditable.

## 3. Objectives

- Provide students, faculty, and administrators with a single unified platform for all academic activities, eliminating the need to switch between multiple disconnected tools.
- Deliver accurate, hallucination-free AI responses by grounding every answer in verified institutional documents through RAG architecture.

- Reduce repetitive administrative and faculty workload by automating tasks such as quiz generation, attendance tracking, grading feedback, and academic announcements.
- Enable real-time institutional visibility for administrators through centralized analytics, early risk detection, and document management.
- Build a permission-aware, role-based AI system where each user — student, faculty, or admin — only accesses information relevant and authorized to their role.

## 4. Methodology / Technology Used

UniGPT is built on a modern full-stack architecture and an AI pipeline designed for reliability and scale:

- RAG Pipeline: User queries and institution-approved documents are converted into vector embeddings using OpenAI's text-embedding-3-large model. Semantic similarity search retrieves the most relevant document chunks, which are then passed as context to GPT-4o for grounded response generation.
- Backend: Laravel 11 (PHP) provides secure REST APIs, role-based access control, and modular service architecture.
- Frontend: Vue.js 3 with Inertia.js and Tailwind CSS delivers a fast, responsive, and role-differentiated user interface.
- Database: MySQL handles structured academic data including users, courses, attendance, assessments, and document metadata.
- AI Models: GPT-4o for reasoning and response generation; text-embedding-3-large for semantic document retrieval.
- AI Chat Modes: Students can switch between academic support, exam preparation, research assistance, assignment help, and career guidance modes.

## 5. Expected Outcomes

Once deployed, UniGPT is expected to deliver the following measurable improvements:

- Significant reduction in the time students spend searching for academic information, as all course materials, policies, and notices become instantly accessible through a single AI-powered interface.
- Decreased faculty workload on repetitive tasks such as quiz creation, grading, and attendance management, allowing more time for quality teaching.
- Higher accuracy and trustworthiness of academic information delivery, as AI responses are grounded in approved documents rather than generic internet data.
- Improved administrative decision-making through real-time dashboards, analytics, and early risk detection for students showing engagement or performance decline.
- A measurable reduction in academic miscommunication, as all official notices, policies, and announcements flow through one verified, centralized platform.

## 6. Core Features

UniGPT offers comprehensive features tailored to three major user groups:

- Students: AI academic copilot with grounded Q&A, multiple chat modes (academic support, exam prep, research, assignment help, career guidance), attendance tracking, GPA/CGPA monitoring, and deadline management.
- Faculty: AI-powered tools for generating quizzes, assignments, and timed class tests; managing attendance; publishing materials; auto-grading assessments; providing AI-assisted feedback; and analyzing course performance.
- Administrators: Centralized control over users, departments, academic structures, document approvals, permissions, announcements, and institution-wide analytics.

## 7. Impact and Feasibility

From a technical perspective, UniGPT is highly feasible and production-ready. The system has already implemented major admin, faculty, and student functionalities using proven, scalable technologies. Its modular architecture supports secure role-based access control, efficient document indexing, and maintainable long-term development.

The societal impact of UniGPT is significant. In a country like Bangladesh where universities often struggle with resource fragmentation and inconsistent academic communication, a centralized AI-powered platform can meaningfully raise the quality and equity of education. By making institutional knowledge instantly accessible to every student and reducing administrative friction for faculty, UniGPT has the potential to improve academic outcomes across institutions of all sizes.

Technologically, UniGPT demonstrates how RAG-based AI can be responsibly deployed in high-stakes academic environments — setting a model for trustworthy, citation-backed AI usage that other institutions can follow.

## 8. Scalability & Future Vision

UniGPT is designed with scalability in mind. Its architecture can support thousands of students, faculty members, departments, and documents across multiple institutions. Future expansion includes real-time messaging, integration with platforms like WhatsApp and Telegram, predictive analytics, voice-based AI interaction, digital library integration, and support for multiple AI providers.

The long-term vision is to transform universities into AI-driven intelligent ecosystems where academic knowledge becomes instantly accessible, administrative friction is minimized, and educational productivity is significantly enhanced.